Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-20 Reference-case-2% - Test 1.experiment_data.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.242 Hz*
Freq. upper bound:	*4.844 Hz*

Freq. Domain Analysis

Overlap 100% (4) series:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*33*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*40*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *48.9%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1207*	*0.05897*
Peak wave period T_P (s):	*1.939*	*2.065*
Energy wave period $T_{-1,0}$ (s):	*1.775*	*1.88*
Mean wave period $T_{0,1}$ (s):	*1.596*	*1.566*
Mean wave period $T_{0,2}$ (s):	*1.481*	*1.343*
Spectral width (Broadness):	*0.748*	*0.8609*
Spectral width (Narrowness):	*0.4008*	*0.6006*

Time Domain Analysis

Overall amplitude reflection coefficient: *48.9%*

	Incident	**Reflected**
Number of waves:	1021	1007
Mean wave height H_m (m):	*0.07742*	*0.03881*
Mean wave period T_m (s):	*1.683*	*1.706*
Sig. wave height H_s (m):	*0.1199*	*0.06117*
$T_{H_{1/3}}$ (s):	*1.887*	*2.028*
H_{max} (m):	*0.1994*	*0.1095*
T_{Hmax} (s):	*1.853*	*2.491*
$H_{1/10}$ (m):	*0.1494*	*0.07771*
$H_{1/50}$ (m):	*0.18*	*0.09118*
$H_{1/100}$ (m)	*0.1885*	*0.09531*
$H_{1/250}$ (m):	*0.1944*	*0.1021*
$H_{0.1\%}$ (m):	*0.1992*	*0.1095*
$H_{1\%}$ (m):	*0.1792*	*0.08904*
$H_{2\%}$ (m):	*0.1684*	*0.08599*
$H_{10\%}$ (m)	*0.1277*	*0.06568*
Groupiness factor GF:	*1.083*	*1.03*
Skewness b1:	*0.6891*	*0.1519*
Kurtosis b2:	*3.43*	*3.151*

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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-20 Reference-case-2% - Test 1.experiment.prof (Prose)*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.242 Hz*
Freq. upper bound:	*4.844 Hz*

Freq. Domain Analysis

Overlap 50% (4) series:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*34*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*40*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *50.3%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1317*	*0.06623*
Peak wave period T_P (s):	*2.065*	*2.065*
Energy wave period $T_{-1,0}$ (s):	*1.823*	*1.963*
Mean wave period $T_{0,1}$ (s):	*1.609*	*1.591*
Mean wave period $T_{0,2}$ (s):	*1.474*	*1.328*
Spectral width (Broadness):	*0.7845*	*0.8814*
Spectral width (Narrowness):	*0.4375*	*0.6598*

Time Domain Analysis

Overall amplitude reflection coefficient: *50.3%*

	Incident	**Reflected**
Number of waves:	1027	1026
Mean wave height H_m (m):	*0.08643*	*0.04427*
Mean wave period T_m (s):	*1.741*	*1.743*
Sig. wave height H_s (m):	*0.1341*	*0.06913*
$T_{H_{1/3}}$ (s):	*1.971*	*2.111*
H_{max} (m):	*0.2145*	*0.1259*
T_{Hmax} (s):	*1.596*	*2.058*
$H_{1/10}$ (m):	*0.1646*	*0.08528*
$H_{1/50}$ (m):	*0.1891*	*0.1*
$H_{1/100}$ (m)	*0.1941*	*0.1058*
$H_{1/250}$ (m):	*0.2019*	*0.1155*
$H_{0.1\%}$ (m):	*0.2142*	*0.1257*
$H_{1\%}$ (m):	*0.1874*	*0.0973*
$H_{2\%}$ (m):	*0.1801*	*0.09215*
$H_{10\%}$ (m)	*0.144*	*0.07449*
Groupiness factor GF:	*1.086*	*0.9722*
Skewness b1:	*0.7965*	*0.1232*
Kurtosis b2:	*3.48*	*2.961*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-20 Reference-case-2% - Test 1.experiment.txt (Phase*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.219 Hz*
Freq. upper bound:	*4.375 Hz*

Freq. Domain Analysis

Overlap 20% (4) series:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*35*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*44*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *51.8%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1455*	*0.07544*
Peak wave period T_P (s):	*2.207*	*2.286*
Energy wave period $T_{-1,0}$ (s):	*1.89*	*2.058*
Mean wave period $T_{0,1}$ (s):	*1.634*	*1.602*
Mean wave period $T_{0,2}$ (s):	*1.478*	*1.303*
Spectral width (Broadness):	*0.809*	*0.8877*
Spectral width (Narrowness):	*0.4728*	*0.7149*

Time Domain Analysis

Overall amplitude reflection coefficient: *52%*

	Incident	**Reflected**
Number of waves:	1027	1017
Mean wave height H_m (m):	*0.09624*	*0.05145*
Mean wave period T_m (s):	*1.797*	*1.817*
Sig. wave height H_s (m):	*0.1479*	*0.07761*
$T_{H_{1/3}}$ (s):	*2.06*	*2.168*
H_{max} (m):	*0.2363*	*0.1411*
T_{Hmax} (s):	*1.695*	*2.07*
$H_{1/10}$ (m):	*0.1784*	*0.09315*
$H_{1/50}$ (m):	*0.1991*	*0.111*
$H_{1/100}$ (m)	*0.2066*	*0.1178*
$H_{1/250}$ (m):	*0.2158*	*0.1291*
$H_{0.1\%}$ (m):	*0.2356*	*0.1409*
$H_{1\%}$ (m):	*0.1971*	*0.1089*
$H_{2\%}$ (m):	*0.1885*	*0.1016*
$H_{10\%}$ (m)	*0.1592*	*0.08089*
Groupiness factor GF:	*1.054*	*0.9243*
Skewness b1:	*0.8254*	*0.108*
Kurtosis b2:	*3.382*	*2.809*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-20 Reference-case-2% - Test 1.experiment.txt (Phase*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.219 Hz*
Freq. upper bound:	*4.375 Hz*

Freq. Domain Analysis

Overlap 30% (4) series:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*37*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*44*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *53%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1561*	*0.08278*
Peak wave period T_P (s):	*2.286*	*2.286*
Energy wave period $T_{-1,0}$ (s):	*1.914*	*2.107*
Mean wave period $T_{0,1}$ (s):	*1.623*	*1.587*
Mean wave period $T_{0,2}$ (s):	*1.45*	*1.265*
Spectral width (Broadness):	*0.8234*	*0.8913*
Spectral width (Narrowness):	*0.5018*	*0.7575*

Time Domain Analysis

Overall amplitude reflection coefficient: *53%*

	Incident	**Reflected**
Number of waves:	1048	1026
Mean wave height H_m (m):	*0.1044*	*0.05759*
Mean wave period T_m (s):	*1.815*	*1.854*
Sig. wave height H_s (m):	*0.1609*	*0.0848*
$T_{H_{1/3}}$ (s):	*2.097*	*2.255*
H_{max} (m):	*0.2336*	*0.1393*
T_{Hmax} (s):	*2.433*	*2.039*
$H_{1/10}$ (m):	*0.1911*	*0.1004*
$H_{1/50}$ (m):	*0.214*	*0.1156*
$H_{1/100}$ (m)	*0.2187*	*0.1197*
$H_{1/250}$ (m):	*0.2232*	*0.1274*
$H_{0.1\%}$ (m):	*0.2332*	*0.1389*
$H_{1\%}$ (m):	*0.2138*	*0.1134*
$H_{2\%}$ (m):	*0.2065*	*0.1094*
$H_{10\%}$ (m)	*0.172*	*0.09023*
Groupiness factor GF:	*1.078*	*0.8812*
Skewness b1:	*0.939*	*0.09602*
Kurtosis b2:	*3.54*	*2.687*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-20 Reference-case-2% - Test 1.experiment.txt (Phase*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.211 Hz*
Freq. upper bound:	*4.219 Hz*

Freq. Domain Analysis

Overlap 40% (4) series:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*38*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*46*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *53.8%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1638*	*0.08817*
Peak wave period T_P (s):	*2.286*	*2.37*
Energy wave period $T_{-1,0}$ (s):	*1.96*	*2.19*
Mean wave period $T_{0,1}$ (s):	*1.627*	*1.581*
Mean wave period $T_{0,2}$ (s):	*1.438*	*1.235*
Spectral width (Broadness):	*0.8358*	*0.8935*
Spectral width (Narrowness):	*0.5287*	*0.7993*

Time Domain Analysis

Overall amplitude reflection coefficient: *53.9%*

	Incident	**Reflected**
Number of waves:	1037	1068
Mean wave height H_m (m):	*0.1128*	*0.06071*
Mean wave period T_m (s):	*1.897*	*1.841*
Sig. wave height H_s (m):	*0.1682*	*0.08908*
$T_{H_{1/3}}$ (s):	*2.213*	*2.311*
H_{max} (m):	*0.2508*	*0.155*
T_{Hmax} (s):	*1.631*	*1.92*
$H_{1/10}$ (m):	*0.1957*	*0.1063*
$H_{1/50}$ (m):	*0.2219*	*0.1234*
$H_{1/100}$ (m)	*0.2297*	*0.1301*
$H_{1/250}$ (m):	*0.2388*	*0.1396*
$H_{0.1\%}$ (m):	*0.2504*	*0.1539*
$H_{1\%}$ (m):	*0.2173*	*0.1221*
$H_{2\%}$ (m):	*0.2115*	*0.1134*
$H_{10\%}$ (m)	*0.1772*	*0.09337*
Groupiness factor GF:	*1.045*	*0.8544*
Skewness b1:	*0.9585*	*0.1094*
Kurtosis b2:	*3.449*	*2.658*

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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-20 Reference-case-2% - Test 1.experiment.50% (4) .txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.211 Hz*
Freq. upper bound:	*4.219 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of adjacent series:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*38*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*46*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *53.6%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1674*	*0.08973*
Peak wave period T_P (s):	*2.37*	*2.37*
Energy wave period $T_{-1,0}$ (s):	*1.981*	*2.232*
Mean wave period $T_{0,1}$ (s):	*1.636*	*1.611*
Mean wave period $T_{0,2}$ (s):	*1.444*	*1.252*
Spectral width (Broadness):	*0.8342*	*0.8959*
Spectral width (Narrowness):	*0.5316*	*0.8086*

Time Domain Analysis

Overall amplitude reflection coefficient: *53.8%*

	Incident	Reflected
Number of waves:	1026	1051
Mean wave height H_m (m):	*0.1154*	*0.06268*
Mean wave period T_m (s):	*1.916*	*1.87*
Sig. wave height H_s (m):	*0.1735*	*0.09023*
$T_{H_{1/3}}$ (s):	*2.229*	*2.323*
H_{max} (m):	*0.2388*	*0.1382*
T_{Hmax} (s):	*2.257*	*2.393*
$H_{1/10}$ (m):	*0.1983*	*0.1073*
$H_{1/50}$ (m):	*0.2191*	*0.1225*
$H_{1/100}$ (m)	*0.2257*	*0.1267*
$H_{1/250}$ (m):	*0.232*	*0.1325*
$H_{0.1\%}$ (m):	*0.2386*	*0.1379*
$H_{1\%}$ (m):	*0.2186*	*0.1208*
$H_{2\%}$ (m):	*0.2099*	*0.1163*
$H_{10\%}$ (m)	*0.183*	*0.09487*
Groupiness factor GF:	*1.066*	*0.8298*
Skewness b1:	*1.022*	*0.08779*
Kurtosis b2:	*3.568*	*2.598*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-20 Reference-case-2% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.328 Hz*
Freq. upper bound:	*6.563 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*26*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*29*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *41%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.08132*	*0.03335*
Peak wave period T_P (s):	*1.524*	*1.524*
Energy wave period $T_{-1,0}$ (s):	*1.44*	*1.442*
Mean wave period $T_{0,1}$ (s):	*1.363*	*1.282*
Mean wave period $T_{0,2}$ (s):	*1.312*	*1.168*
Spectral width (Broadness):	*0.6038*	*0.7874*
Spectral width (Narrowness):	*0.2797*	*0.4537*

Time Domain Analysis

Overall amplitude reflection coefficient: *41%*

	Incident	**Reflected**
Number of waves:	1009	1052
Mean wave height H_m (m):	*0.05102*	*0.02115*
Mean wave period T_m (s):	*1.355*	*1.3*
Sig. wave height H_s (m):	*0.07829*	*0.03349*
$T_{H_{1/3}}$ (s):	*1.485*	*1.528*
H_{max} (m):	*0.1372*	*0.05467*
T_{Hmax} (s):	*1.585*	*1.592*
$H_{1/10}$ (m):	*0.09533*	*0.04164*
$H_{1/50}$ (m):	*0.1121*	*0.04922*
$H_{1/100}$ (m)	*0.1194*	*0.05169*
$H_{1/250}$ (m):	*0.1274*	*0.05344*
$H_{0.1\%}$ (m):	*0.1372*	*0.05463*
$H_{1\%}$ (m):	*0.1128*	*0.04924*
$H_{2\%}$ (m):	*0.102*	*0.04531*
$H_{10\%}$ (m)	*0.08294*	*0.0364*
Groupiness factor GF:	*0.9768*	*1.013*
Skewness b1:	*0.3184*	*0.2992*
Kurtosis b2:	*2.951*	*3.088*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-20 Reference-case-2% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.281 Hz*
Freq. upper bound:	*5.625 Hz*

Freq. Domain Analysis

Overlap of (4) subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*30*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*34*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *45.8%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.108*	*0.04941*
Peak wave period T_P (s):	*1.778*	*1.778*
Energy wave period $T_{-1,0}$ (s):	*1.621*	*1.672*
Mean wave period $T_{0,1}$ (s):	*1.495*	*1.432*
Mean wave period $T_{0,2}$ (s):	*1.413*	*1.261*
Spectral width (Broadness):	*0.7021*	*0.8344*
Spectral width (Narrowness):	*0.3476*	*0.5384*

Time Domain Analysis

Overall amplitude reflection coefficient: *45.9%*

	Incident	**Reflected**
Number of waves:	1026	1017
Mean wave height H_m (m):	*0.0674*	*0.03223*
Mean wave period T_m (s):	*1.512*	*1.525*
Sig. wave height H_s (m):	*0.1047*	*0.05088*
$T_{H_{1/3}}$ (s):	*1.691*	*1.77*
H_{max} (m):	*0.1857*	*0.1047*
T_{Hmax} (s):	*1.695*	*1.824*
$H_{1/10}$ (m):	*0.1289*	*0.0644*
$H_{1/50}$ (m):	*0.155*	*0.07995*
$H_{1/100}$ (m)	*0.1638*	*0.08646*
$H_{1/250}$ (m):	*0.1765*	*0.0955*
$H_{0.1\%}$ (m):	*0.1855*	*0.1046*
$H_{1\%}$ (m):	*0.1496*	*0.07784*
$H_{2\%}$ (m):	*0.1419*	*0.06978*
$H_{10\%}$ (m)	*0.1114*	*0.05484*
Groupiness factor GF:	*1.037*	*1.062*
Skewness b1:	*0.5437*	*0.1952*
Kurtosis b2:	*3.162*	*3.247*